



OpenSolaris on Xen **Part 1**

The Ecosystem of the Xen Platform

This article deals with the concepts associated with Xen virtualisation technology—something we should be aware of before we get started with Xen deployments.

Being able to run multiple operating systems concurrently on the same computer is gaining currency, of late, since this reduces costs and increases utilisation of systems. Being able to carve multiple virtual machines (VM) out of a single physical machine is called virtualisation.

Virtualisation of the underlying platform can be achieved in multiple ways. For example, VirtualBox takes an unmodified OS and runs it as a process inside another OS. Another approach is to run a thin layer of software called a hypervisor on the bare metal and make all the guests run on top of the hypervisor. Xen adopts this model.

It uses para-virtualisation (PV) to boost the performance of the VMs. To run as a PV guest, an OS needs to be ported to the Xen architecture—which is very similar to the actual underlying platform (i.e., the x86/x64 architecture). Thus, a PV guest is aware that it is running on top of a hypervisor and hence not in control of the underlying physical machine.

In this four-article series we shall explore: 1) The ecosystem of an OpenSolaris Xen platform and other under-the-hood machinations of the hypervisor; 2) OpenSolaris as dom0; 3) The changing architecture of Xen in the face of new hardware advances; and 4) A high level code walk-through of the dom0 kernel.